

Climate 5000 Series - Gen 4

18K BTU Single Zone Regular System

1-Way Cassette IDU

Engineering
Submittal
Sheet



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Overview and Project Information



Project name

Location

Architect

Engineer

General contractor

Reference #

18K Btu Single Zone Regular System - 1-Way Cassette IDU

Indoor part number	8-733-962-668
Indoor model number	BMS500-AAU018-1AHZXD
Outdoor part number	8-733962-686
Outdoor model number	BMS500-AAS018-1CSXRD

Efficiency Data		AHRI 210/240 2023
SEER2		22.2
EER2		12.5
HSPF2 Region IV		12.2
HSPF2 Region V		8.6
COP(47F)		3.30
COP (5F)		2.24

Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	4000-18800
	Rated Capacity	18000
	Current	6.4
	Input Power	1440
Heating	Capacity Range	7600-22800
	Rated Capacity (47F)	18000
	Rated Capacity (5F)	13100
	Current	7.1
	Input Power	1600

Electrical Data	
Power supply (V / Ø / Hz)	208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)	3.0 / 18.0
Max. fuse indoor/outdoor (A)	15.0 / 20.0

Compressor Data	
Type:	Rotary
Refrigerant oil /oil charge (ml)	VG74 / 440
Input (W)	1090
Rated current (RLA/A)	7.2

Piping	
Refrigerant precharge (oz/kg)	47.97 / 1.36
Length for precharge (ft/m)	24.6 / 7.5
Liquid side (in/mm)	1/4" / Ø6.35
Gas side (in/mm)	1/2" / Ø12.7
Additional Charge of Refrigerant (oz/ft)/(g/m)	0.16 / 15
Max. refrigerant pipe length (ft/m)	98.40 / 30
Max. difference in height (ft/m)	65.60 / 20

Operation Range

ODU	Cooling (°F / °C)	-13~122 / -25~50
	Heating (°F / °C)	-13~75 / -25~24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

System Data

Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		4.40
IDU	Fan motor model	ZKFN-30-8-43
	Fan motor qty (ea)	1
	Fan motor input (W)	46.0
	Fan speed (RPM)	1344 / 1184 / 1024
	Air flow, Turbo / H / M / L / Si (CFM)	400.2 / 352.0 / 300.2 / 207.2 / N/A
	Approx. Air Throw Range ft (m)	24.6 / 7.5
	Noise level, Turbo / H / M / L / Si (dB(A))	N/A / 44.0 / 41.0 / 29.5 / 24.0
	Net weight (lbs / kg)	45.19 / 20.5
ODU	Gross weight (lbs / kg)	82.89 / 37.6
	Drain pipe diameter (mm)	ODΦ25mm
	Fan motor model	ZKFN-80-10-1L
	Fan motor qty (ea)	1
	Fan motor input (W)	161
	Fan motor speed (RPM)	780 / 700 / 550
	Air flow, Max (CFM)	1765.8
	Noise level, Max (dB(A))	57.5
	Net weight (lbs / kg)	92.59 / 42.0
	Gross weight (lbs / kg)	100.31 / 45.5

Accessories

BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
BMS-DRY-XXD (8733969263)	24V Interface
BMS-PK2-7X1 (8733951011)	25ft Lineset

Standard Parts

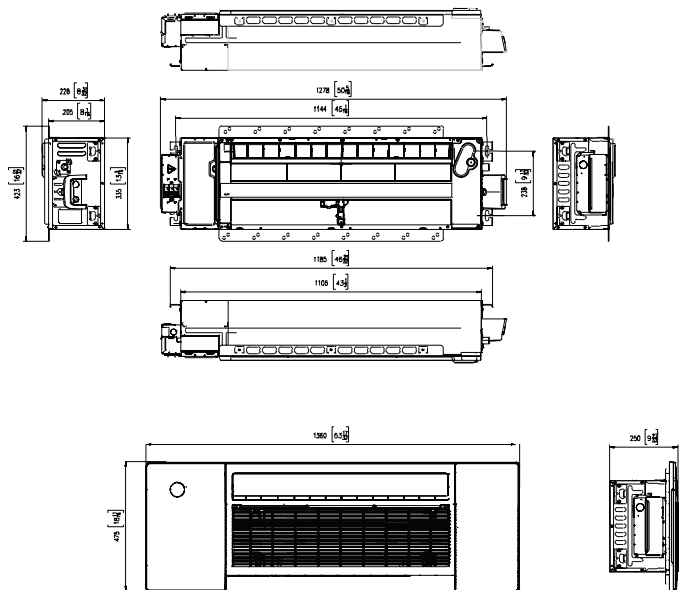
8-733-969-065	Remote Control
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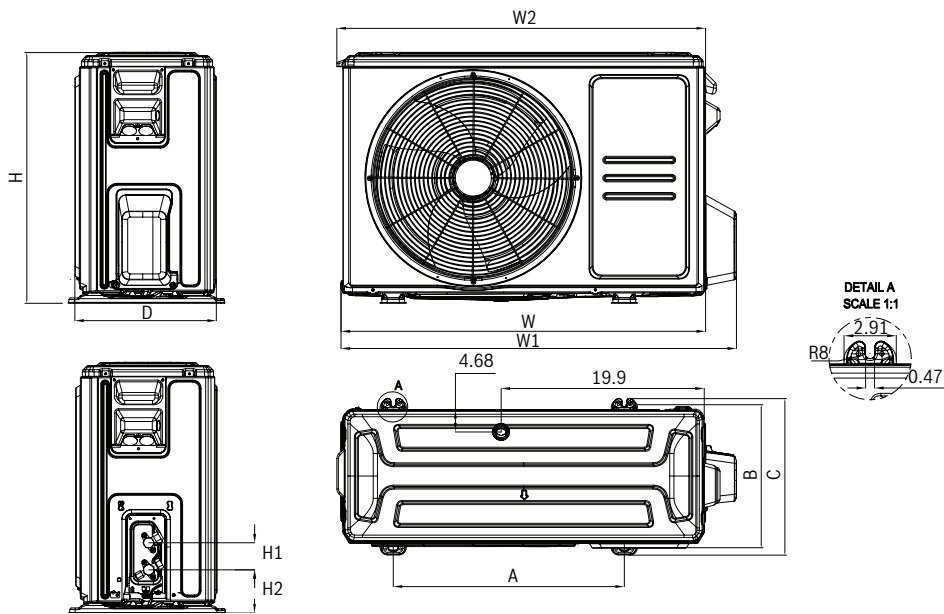


Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU018-1AHZXD	50.31	13.19	8.98

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	A (in)	B (in)	C (in)	H1 (in)	H2 (in)
BMS500-AAS018-1CSXRD	35.0	13.46	26.5	37.6	35.24	26.1	13.7	14.96	2.36	4.25

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Cooling Performance - Indoor Dry Bulb Temperature at 400 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-22°F (-30.0°C)	TC	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/
	S/T	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/
	PI	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/
-13°F (-25.0°C)	TC	14550	14543	14751	14812	15219	15181	15168	15158	15157	15145	15634	15615	15600	15598	15583	16491	16487	16453	16447	16412
	S/T	0.80	0.91	1.00	1.00	1.00	0.65	0.75	0.85	0.86	0.97	0.55	0.65	0.74	0.76	0.86	0.39	0.47	0.56	0.57	0.67
	PI	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66
-4°F (-20.0°C)	TC	15669	15662	15885	15951	16390	16348	16335	16324	16323	16309	16837	16816	16800	16798	16782	17760	17755	17719	17712	17674
	S/T	0.80	0.92	1.00	1.00	1.00	0.65	0.76	0.85	0.87	0.98	0.55	0.65	0.74	0.76	0.86	0.39	0.48	0.56	0.57	0.67
	PI	0.68	0.69	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68
0°F (-17.8°C)	TC	16189	16179	16480	16548	17003	16883	16865	16852	16850	17019	17380	17355	17333	17331	17322	18314	18310	18308	18311	18323
	S/T	0.81	0.93	1.00	1.00	1.00	0.66	0.77	0.86	0.88	1.00	0.55	0.66	0.75	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.70	0.70	0.70	0.70	0.70
5°F (-15.0°C)	TC	16653	16710	17092	17171	17700	17434	17439	17440	17458	17718	17992	17996	18000	18000	18024	19047	19041	19051	19053	19062
	S/T	0.82	0.94	1.00	1.00	1.00	0.66	0.78	0.87	0.88	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.59	0.69
	PI	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.72	0.72	0.72	0.72	0.72
14°F (-10.0°C)	TC	17326	17317	17671	17753	18304	18186	18170	18158	18156	18326	18809	18786	18766	18763	18745	20000	19994	19955	19949	19906
	S/T	0.81	0.93	1.00	1.00	1.00	0.65	0.76	0.86	0.87	1.00	0.55	0.66	0.75	0.76	0.87	0.39	0.48	0.56	0.57	0.67
	PI	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63
17°F (-8.3°C)	TC	17508	17566	17971	18059	18646	18380	18388	18391	18406	18666	19007	19014	19022	19023	19046	20215	20209	20206	20210	20227
	S/T	0.82	0.93	1.00	1.00	1.00	0.66	0.77	0.87	0.88	1.00	0.55	0.66	0.76	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.61	0.61	0.61	0.61	0.61
23°F (-5.0°C)	TC	17259	17333	17781	17869	18458	18133	18138	18140	18164	18478	18758	18765	18769	18770	18800	19946	19939	19954	19956	19968
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.89	0.39	0.48	0.57	0.59	0.69
	PI	0.63	0.63	0.63	0.63	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.65	0.65	0.65	0.65	0.65	0.65
35°F (1.7°C)	TC	16919	16920	17019	17067	17617	17710	17710	17711	17758	18265	18265	18264	18265	18267	19293	19289	19288	19287	19288	19288
	S/T	0.78	0.88	0.97	0.99	1.00	0.63	0.73	0.82	0.83	0.93	0.54	0.63	0.72	0.73	0.83	0.39	0.46	0.54	0.56	0.65
	PI	0.65	0.65	0.65	0.65	0.68	0.68	0.68	0.68	0.68	0.69	0.71	0.71	0.71	0.71	0.71	0.78	0.78	0.78	0.78	0.78
41°F (5.0°C)	TC	16506	16506	16638	16753	17248	17274	17273	17274	17335	17813	17811	17812	17813	17814	18812	18807	18806	18805	18807	18807
	S/T	0.78	0.89	0.98	1.00	1.00	0.64	0.74	0.83	0.84	0.94	0.54	0.64	0.72	0.74	0.84	0.39	0.47	0.55	0.56	0.66
	PI	0.68	0.68	0.69	0.70	0.73	0.73	0.73	0.73	0.73	0.76	0.76	0.76	0.76	0.76	0.83	0.83	0.83	0.83	0.83	0.83
47°F (8.3°C)	TC	16092	16102	16318	16391	16876	16838	16837	16838	16837	16915	17361	17359	17360	17360	18333	18328	18326	18326	18327	18327
	S/T	0.79	0.90	1.00	1.00	1.00	0.64	0.75	0.84	0.85	0.96	0.54	0.65	0.73	0.75	0.85	0.39	0.47	0.55	0.57	0.67
	PI	0.72	0.72	0.74	0.74	0.77	0.77	0.77	0.77	0.77	0.78	0.81	0.81	0.81	0.81	0.88	0.88	0.88	0.88	0.88	0.88
50°F (10.0°C)	TC	15899	16337	16888	16986	17644	16797	16806	16922	16975	17661	17452	17474	17484	17485	17653	18655	18704	18752	18758	18781
	S/T	0.90	1.00	1.00	1.00	1.00	0.71	0.85	0.96	0.99	1.00	0.59	0.72	0.83	0.85	0.99	0.40	0.50	0.60	0.62	0.74
	PI	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.83	0.82	0.82	0.82
59°F (15.0°C)	TC	16184	16752	17346	17452	18162	17123	17147	17341	17470	18180	17830	17848	17854	17880	18195	19145	19184	19222	19226	19242
	S/T	0.92	1.00	1.00	1.00	1.00	0.73	0.86	1.00	1.00	1.00	0.60	0.73	0.85	0.86	1.00	0.40	0.51	0.61	0.63	0.76
	PI	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.88	0.89	0.89	0.89	0.89	0.89
68°F (20.0°C)	TC	16459	17161	17798	17911	18671	17424	17464	17817	17930	18690	18178	18191	18224	18240	18705	19586	19623	19654	19657	19667
	S/T	0.95	1.00	1.00	1.00	1.00	0.74	0.89	1.00	1.00	1.00	0.61	0.75	0.87	0.89	1.00	0.40	0.51	0.63	0.65	0.78
	PI	0.93	0.93	0.94	0.94	0.95	0.94	0.94	0.94	0.94	0.95	0.94	0.94	0.94	0.94	0.95	0.96	0.95	0.95	0.95	0.95
77°F (25.0°C)	TC	16718	17523	18209	18330	19149	17723	17780	18229	18351	19168	18526	18547	18594	18615	19185	20009	20070	20116	20121	20140
	S/T	0.97	1.00	1.00	1.00	1.00	0.75	0.90	1.00	1.00	1.00	0.61	0.76	0.88	0.90	1.00	0.40	0.52	0.63	0.65	0.79
	PI	0.99	0.99	1.00	1.00	1.00	0.99	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.01	1.01	1.01	1.01
82°F (27.8°C)	TC	16828	17687	18400	18525	19373	17832	17901	18416	18546	19394	18662	18685	18743	18767	19411	20177	20253	20306	20311	20333
	S/T	0.99	1.00	1.00	1.00	1.00	0.76	0.91	1.00	1.00	1.00	0.62	0.77	0.89	0.92	1.00	0.40	0.52	0.64	0.66	0.80
	PI	1.09	1.10	1.10	1.10	1.11	1.10	1.10	1.10	1.10	1.11	1.10	1.10	1.10	1.10	1.11	1.12	1.11	1.11	1.11	1.11
86°F (30.0°C)	TC	16931	17776	18503	18632	19500	17855	17937	18524	18654	19520	18701	18723	18792	18819	19537	20246	20325	20373	20378	20397
	S/T	1.00	1.00	1.00	1.00	1.00	0.77	0.93	1.00	1.00	1.00	0.62	0.78	0.91	0.93	1.00	0.40	0.52	0.64	0.66	0.81
	PI	1.26	1.27	1.27	1.27	1.28	1.27	1.27	1.27	1.27	1.28	1.27	1.27	1.27	1.27	1.28	1.29	1.29	1.29	1.29	1.29
95°F (35.0°C)	TC	17038	17130	17683	17788	18491	18107	18118	18124	18145	18514	18884	18888	18900	18902	18933	20349	20341	20345	20351	20378
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76							

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Heating Performance - Indoor Dry Bulb Temperature at 400 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	/	/	/	/	/	/	/	/	/	/
-13°F(-25.0°C)	8307	8141	8100	8068	7995	1.48	1.49	1.49	1.50	1.50
-4°F(-20.0°C)	10563	10352	10300	10259	10166	1.55	1.56	1.56	1.57	1.57
0°F(-17.8°C)	11839	11602	11544	11498	11395	1.61	1.63	1.63	1.63	1.64
5°F(-15.0°C)	13435	13166	13100	13048	12930	1.70	1.71	1.71	1.72	1.72
14°F(-10.0°C)	15050	14749	14675	14616	14485	1.76	1.77	1.78	1.78	1.79
17°F(-8.3°C)	15588	15276	15200	15139	15003	1.78	1.79	1.80	1.80	1.81
22°F(-5.6°C)	16432	16103	16023	15959	15815	1.82	1.83	1.84	1.84	1.85
27°F(-2.8°C)	17276	16930	16846	16778	16627	1.86	1.87	1.88	1.88	1.89
32°F(0.0°C)	18120	17757	17669	17598	17440	1.90	1.91	1.92	1.92	1.93
35°F(1.7°C)	18626	18254	18162	18090	17927	1.92	1.94	1.94	1.95	1.95
37°F(2.8°C)	19419	19031	18935	18860	18690	1.91	1.93	1.93	1.94	1.94
42°F(5.6°C)	21401	20973	20868	20784	20597	1.89	1.90	1.91	1.91	1.92
44.6°F(7.0°C)	22193	21749	21641	21554	21360	1.88	1.89	1.90	1.90	1.91
47°F(8.3°C)	23382	22915	22800	22709	22504	1.86	1.88	1.88	1.88	1.89
52°F(11.1°C)	22361	21914	21805	21717	21522	1.65	1.66	1.66	1.67	1.67
57°F(13.9°C)	21340	20914	20809	20726	20539	1.43	1.44	1.44	1.45	1.45
59°F(15.0°C)	20932	20513	20411	20329	20146	1.34	1.35	1.36	1.36	1.37
62°F(16.7°C)	20320	19913	19814	19734	19557	1.21	1.22	1.23	1.23	1.23
64.4°F(18.0°C)	19830	19433	19336	19258	19085	1.11	1.12	1.12	1.12	1.13
64.4°F(18.0°C)	18727	18352	18261	18188	18024	0.88	0.88	0.89	0.89	0.89
75.2°F(24.0°C)	17624	17272	17186	17117	16963	0.65	0.65	0.65	0.65	0.66

Bolded values were tested. All other values are simulated and for reference only.